

Karneeshkar V

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[Portfolio](#) | [Terminal portfolio](#)

PROFESSIONAL SUMMARY

Embedded software engineer delivering real-time IoT and automotive systems from prototype to production. Proficient in C/C++/Rust/Python on Zephyr RTOS and QNX/Linux across STM32/ESP32; experienced with LoRaWAN, MQTT, and automotive diagnostics for connected devices. Skilled in edge AI, sensor fusion, and performance optimization for reliable, resource-constrained platforms.

TECHNICAL SKILLS

Languages: Rust, C/C++, Python, Bash

Embedded & IoT: Zephyr RTOS, Espressif IDF, STM32, ESP32, Arduino, Raspberry Pi, LoRaWAN, MQTT

AI & Data: PyTorch, TensorFlow, Computer Vision, Sensor Data Processing

Microcontrollers & Tools: ESP32, Pi Pico, JTAG, STM32

Soft skills: Communication, Teamwork, Active listening, Problem solving, Critical thinking, Time management

EXPERIENCE

Software Developer 2

Jul 2025 – Current

2Cents Capital

Bengaluru, India

- Led and scaled cross-functional Mobile and AI teams (10+ engineers) to deliver high-impact features with 95% on-time release rate.
- Published a fully responsive investment application on both iOS App Store and Google Play, achieving <0.5s cold-start time.
- Designed and deployed AI agents across the end-to-end customer journey—including customer support, onboarding, and market research—improving support resolution speed by 40% and reducing manual intervention costs by 30%.

Software Developer

Jan 2025–Jun 2025

Visteon

Chennai, India

- Created diagnostic routines for JLR platforms that detect system anomalies with 96% accuracy, boosting vehicle reliability.
- Optimized real-time dashboard software (SomeIP on QNX/Linux) in C, reducing processing latency by 25%.
- Wrote unit and functional tests to reach 99.8% code coverage across the code-base with zero deviations.

Automation Intern

Dec 2024–Jan 2025

Procter & Gamble

Hyderabad, India

- Implemented IIoT monitoring on five production lines, ingesting 2M+ data points daily with 99.9% uptime.
- Automated quality-control workflows, cutting manual inspection time 40% and improving defect detection 15%.

IoT & LoRaWAN Developer

May 2024–Jun 2024

Tynatech Ingenious Pvt Ltd

Noida, India

- Built RESTful APIs/webhooks connecting 50+ LoRaWAN devices to cloud platforms.
- Optimized UART–LoRaWAN interface to raise throughput 30% via efficient buffer management.
- Added MQTT streaming for real-time gateway visualization, handling 100+ msgs s⁻¹.

Web Development Intern

Oct 2023–Dec 2023

Intellect Design Arena

Chennai, India

- Developed PHP-based financial portals with 30% faster page loads.
- Managed cloud infrastructure for five web apps, configuring auto-scaling for 99.9% availability.
- Created analytics dashboards delivering real-time engagement and conversion insights.

R&D Intern

Aug 2023–Sep 2023

Hindustan Aeronautics Ltd (HAL)

Bangalore, India

- Optimized autopilot algorithms, cutting computational overhead 22% while meeting safety constraints.
- Authored C++ simulation tool covering 200+ control-system test scenarios.
- Assisted avionics teams with system documentation and verification protocols.

PROJECTS

- Multipurpose VR Gloves with Gesture Recognition** | *Unity, C#, ESP32, ML* Sep 2024
- Built VR gloves capturing 15+ gestures at 92% accuracy using flex-sensor ML pipelines.
 - Developed a web UI to map gestures to 20+ system commands; integrated Unity for gaming/assistive control.
- Embedded Athlete-Performance Monitoring System** | *ESP32, PyTorch, OpenCV* Mar 2024
- Ran edge CV on ESP32 to analyze runner form in real time; fused MPU6050, VO₂, and temperature signals.
 - Built a decision-tree classifier with a Flask/Plotly dashboard for actionable feedback.
- Smart Rhesus Macaque Deterrent System** | *Jetson Nano, CV, Ultrasonic* Mar 2024
- Achieved 94% macaque detection in varied lighting; ultrasonic emitters cut intrusions 90%.
 - Power-management module lowered energy use 35% for solar deployment.
- Wearable Jump-Height Measurement Device** | *ESP32, MPU6050, Flask* Mar 2024
- Ankle-mounted sensor reached ± 1 cm accuracy vs lab force-plates.
 - Used Kalman filtering to suppress sensor noise 80%; built SQLite/Flask analytics dashboard.
- RFID-based Runner Tracking System** | *Industrial RFID, PHP, Raspberry Pi* Dec 2023
- Automated timing for 5 km training runs via industrial RFID readers.
 - Web interface enabled seamless record management and legacy-system integration.

EDUCATION

Vellore Institute of Technology Chennai, India
B.Tech. Electronics & Computer Engineering; GPA: 8.66/10